THE UNIVERSITY OF MELBOURNE

Student number

Semester 2 Final Subject Assessment, 2022

School of Mathematics and Statistics

MAST20026 Real Analysis

Reading time: 30 minutes — Writing time: 3 hours — Upload time: 30 minutes

This exam consists of 28 pages (including this page) with 16 questions and 72 total marks

Permitted Materials

- This exam and/or an offline electronic PDF reader, one or more copies of the masked exam template made available earlier and blank loose-leaf paper.
- One double-sided handwritten A4 sheet
- No calculators are permitted. No headphones or earphones are permitted.

Instructions to Students

- Wave your hand right in front of your webcam if you wish to communicate with the supervisor at any time (before, during or after the exam).
- You must not be out of webcam view at any time without supervisor permission.
- You must not write your answers on an iPad or other electronic device.
- Off-line PDF readers (i) must have the screen visible in Zoom; (ii) must only be used to read exam questions (do not access other software or files); (iii) must be set in flight mode or have both internet and Bluetooth disabled as soon as the exam paper is downloaded.

Writing

- If you are writing answers on the exam or masked exam and you need more space, use blank paper. Note this in the answer box, so the marker knows.
- If you are only writing on blank A4 paper, the first page must contain only your student number, subject code and subject name. Write on one side of each sheet only. Start each question on a new page and include the question number at the top of each page.

Scanning and Submitting

- **You must not leave Zoom supervision to scan your exam.** Put the pages in number order and the correct way up. Add any extra pages to the end. Use a scanning app to scan all pages to PDF. Scan directly from above. Crop pages to A4.
- Submit your scanned exam as a single PDF file and carefully review the submission in Gradescope. Scan again and resubmit if necessary. Do not leave Zoom supervision until you have confirmed orally with the supervisor that you have received the Gradescope confirmation email.
- **You must not submit or resubmit after having left Zoom supervision.**

Page 1 of 28 — add any extra pages after page 28 — Page 1 of 28

Page 2 of 28 — add any extra pages after page 28 — Page 2 of 28

Question 1 (3 marks)

Using the definition of logical equivalence determine if the following statement is true:

$$\sim(p \Rightarrow q) \equiv (p \wedge \sim q).$$

Write a concluding sentence that explains how the work you have done supports your conclusion.

p	q	$p \Rightarrow q$	$\sim (p \Rightarrow q)$	$\sim q$	$p \wedge \sim q$
T	T	T	F	F	F
T	F	F	T	T	T
F	T	T	F	F	F
F	F	T	F	T	T

$\sim(p \Rightarrow q) \iff (p \wedge \sim q)$
T
T
T
T

Since $\sim(p \Rightarrow q) \iff (p \wedge \sim q)$ is a tautology we have

$$\sim(p \Rightarrow q) \equiv (p \wedge \sim q),$$

and so the statement is true.

Page 3 of 28 — add any extra pages after page 28 — Page 3 of 28

Question 2 (6 marks)

(a) Let (f_n) be a sequence. Express the following as statement in first-order logic:
If is monotone increasing, then it is bounded.

$$[(\forall n, m \in \mathbb{N})[(n > m) \Rightarrow (f_n \geq f_m)]] \Rightarrow [(\exists N)[(\forall n \in \mathbb{N})(|f_n| < N)]]$$

(b) Express the following as statement in first-order logic:
Every real number has an additive inverse.

$$(\forall r \in \mathbb{R})[(\exists r' \in \mathbb{R}) \;\; r + r' = 0]$$

(c) Let A be a subset of $\mathbb{R}$. Express the following as statement in first-order logic:
Every subset of A has a least element.

$$(\forall S \subset A)[(\exists a \in S)[(\forall x \in S)(x \leq a)]]$$

There is extra space for Question 2 on the subsequent page

Extra space for Question 2. Submit this page, even if left blank.

Page 4 of 28 — add any extra pages after page 28 — Page 4 of 28

Page 5 of 28 — add any extra pages after page 28 — Page 5 of 28

Question 3 (4 marks)

Let C_1 and C_2 be cuts and consider the following statement:

$$(C_1 \cup C_2) \setminus (C_1 \cap C_2)$$

is a cut. Determine if the statement is true. If the statement is true, provide a proof. Otherwise, provide a counterexample that proves the statement is false.

False. Let C_1 ad C_2 be cuts with $C_1 = C_2$. Then, $(C_1 \cup C_2) = (C_1 \cap C_2)$, and hence $(C_1 \cup C_2) \setminus (C_1 \cap C_2) = \emptyset$. This violates condition 3.

Page 6 of 28 — add any extra pages after page 28 — Page 6 of 28

Question 4 (3 marks)

Let A be a natural number, and $B \subseteq A$. Is B a natural number? Briefly justify your response, or give a counterexample.

No. $2 = \{\emptyset, \{\emptyset\}\}$ is a natural number, but and $A := \{\{\emptyset\}\}$, a subset of 2, is not a natural number (it violates condition 1).

Question 5 (3 marks)

Let A be the cut, and $q \in A$. Is $(q)_{\mathbf{R}^\star} \subseteq A$? Briefly justify your response, or give a conterexample.

Yes. Since $q \in A$, $q \in \mathbb{Q}$. Hence $(q)_{\mathbf{R}^\star} = \{r \in \mathbb{Q} \mid r < q\}$. To show $(q)_{\mathbf{R}^\star} \subseteq A$, let $r \in (q)_{\mathbf{R}^\star}$. We have $r < q$. Then, $r \in A$ due to Condition 1 of a cut.

Page 7 of 28 — add any extra pages after page 28 — Page 7 of 28

Question 6 (4 marks)

For each of the following, state whether there exists a sequence that satisfies the given criteria. For those where the sequence exists, give an example. For those where the sequence does not exist, write `there is no such sequence`.

(a) (f_n) does not converge and is bounded.

$$f_n = (-1)^n$$

(b) (f_n) is a Cauchy sequence, and is unbounded.

There is no such sequence

(c) (f_n) converges and has a divergent subsequence.

There is no such sequence.

(d) (f_n) is monotonic and diverges.

$$f_n = n.$$

There is extra space for Question 6 on the subsequent page

Extra space for Question 6. Submit this page, even if left blank.

Page 8 of 28 — add any extra pages after page 28 — Page 8 of 28

Page 9 of 28 — add any extra pages after page 28 — Page 9 of 28

Question 7 (5 marks)

Let $L \in \mathbb{R}$. Let (f_n) and (g_n) be sequences such that

$$(f_n) \to L \qquad \text{and} \qquad (g_n) \to H.$$

and $L < H$. Prove there is an $M \in \mathbb{N}$ such that $f_n < g_n$ for all $n > M$.

Let $\epsilon = \frac{H-L}{2}$, which is positive. By definition of limit, there is an $M_1 \in \mathbb{N}$ so that $f_n < L + \epsilon$ for all $n > M_1$; there is an $M_2 \in \mathbb{N}$ so that $H - \epsilon < g_n$ for all $n > M_2$. Let $M = \max M_1, M_2$. Then, for all $n > M$, we have $f_n < L + \epsilon = H - \epsilon < g_n$.

Page 10 of 28 — add any extra pages after page 28 — Page 10 of 28

Question 8 (5 marks)

Using an ϵ-δ proof, prove

$$\lim_{x \to 2} \frac{x^2}{x-1} = 4.$$

Use this page for any scratch work. Write your proof on the subsequent page. Hand in this page as part of your GradeScope submission as work you do on this page may be used to award partial credit. In your proof you may refer back to any computations you do on this page. In doing so, please make clear where to find the relevant computation on this page.

Page 11 of 28 — add any extra pages after page 28 — Page 11 of 28

Let $\epsilon > 0$. Let $\delta = \min\{1/2, \sqrt{\frac{\epsilon}{2}}\}$.
Notice that x satisfies $0 < |x-2| < \frac{1}{2}$ and hence we have

$$|x-1| > 1/2 \text{ and } \frac{1}{|x-1|} < 2.$$

We also have $0 < |x-1| < \sqrt{\frac{\epsilon}{2}}$, hence

$$\begin{aligned}\left|\frac{x^2}{x-1} - 4\right| &= \left|\frac{(x-2)^2}{x-1}\right| \\ &< 2(\sqrt{\frac{\epsilon}{2}})^2 \\ &= \epsilon\end{aligned}$$

Therefore

$$\lim_{x\to 2} \frac{x^2}{x-1} = 4.$$

Page 12 of 28 — add any extra pages after page 28 — Page 12 of 28

Question 9 (5 marks)

Prove the following theorem:
Let $E \subseteq \mathbb{R}$ and let $a \in \mathbb{R}$. If there exists a sequence $f : \mathbb{N}^+ \to E \setminus \{a\}$ such that $f_n \to a$, then a is a limit point E.

Let $\delta > 0$. By definition of limit, there is an $M \in \mathbb{N}$ so that $|f_n - a| < \delta$ for all $n > M$. Take $n = M + 1$, then we have $|f_n - a| < \delta$. Since $f_n \in E \setminus \{a\}$, we also have $0 < |f_n - a|$. Therefore, we found an element $f_n \in E$ with $0 < |f_n - a| < \delta$.

Page 13 of 28 — add any extra pages after page 28 — Page 13 of 28

Question 10 (7 marks)

Let $g : [-1, 1] \to \mathbb{R}$ such that g is continuous and $0 < g(x) < 1$ for all $x \in [-1, 1]$.

(a) Prove that there is an $\epsilon > 0$ so that $\epsilon < g(x) < 1 - \epsilon$ for all $x \in [-1, 1]$.

(b) Let $(f_n(x))$ be a sequence of functions that converges uniformly to g on $[-1, 1]$. Prove there exists $M \in \mathbb{N}^+$ such that for all $x \in [-1, 1]$ and all $n > M$ the following inequality holds:

$$0 < f_n(x) < 1.$$

Please use the the following space for both Part(a) and Part(b). There is extra space for Question 10 on the subsequent page

(a) By the Extreme Value Theorem, g attains a minimum and a maximum on $[-1, 1]$. Let c and C respectively be these values. We have $0 < c, C < 1$. Let $\epsilon = \min c/2, (1 - C)/2$. Then,

$$\epsilon < c \leq g(x) \leq C < 1 - \epsilon$$

for all $x \in [-1, 1]$.

(b) We have by construction in (a), $\epsilon < c$ and $\epsilon < 1 - C$. Since $(f_n(x))$ converges uniformly to g on $[-1, 1]$, there exists $M \in \mathbb{N}^+$ such that for all $x \in [-1, 1]$ we have $|f_n(x) - g(x)| < \epsilon$ whenever $n > M$. Unpacking the absolute value, we have

$$g(x) - \epsilon < f_n(x) < g(x) + \epsilon$$

Since $c \leq g(x) \leq C$ we have

$$c - \epsilon < f_n(x) < \epsilon + C$$

By construction, $\epsilon < 1 - C$ and $-c < -\epsilon$. Therefore

$$0 = c - c < c - \epsilon < f_n(x) < \epsilon + C < 1 - C + C = 1$$

and so

$$0 < f_n(x) < 1$$

for all $x \in [-1, 1]$ whenever $n > M$.

There is extra space for Question 10 on the subsequent page

Extra space for Question 10. Submit this page, even if left blank.

Page 14 of 28 — add any extra pages after page 28 — Page 14 of 28

Page 15 of 28 — add any extra pages after page 28 — Page 15 of 28

Question 11 (5 marks)

Let $f, g : \mathbb{R} \to \mathbb{R}$ be differentiable on $\mathbb{R}$ such that $f'(x) > g'(x)$ for all $x \in \mathbb{R}$, and $f(0) = g(0)$. Prove that $f(x) > g(x)$ for all $x > 0$.

Let $h(x) = f(x) - g(x)$. Then $h(x)$ is differentiable on $\mathbb{R}$, and $h'(x) = f'(x) - g'(x) > 0$ for all $x \in \mathbb{R}$. We also have $h(0) = 0$. Let $x > 0$. Using the Mean Value Theorem, there is an $0 < c < x$ such that $h'(c) = \frac{h(x)}{x}$. Then, $h'(c) > 0$ implies that $\frac{h(x)}{x} > 0$. Using $x > 0$, we have $h(x) > 0$ and therefore $f(x) > g(x)$.

Page 16 of 28 — add any extra pages after page 28 — Page 16 of 28

Question 12 (3 marks)

Consider the following integrable function

$$f(x) = \begin{cases} -x & x < 0 \\ x^2 & x \geq 0. \end{cases}$$

Use the partition $P = \{-2, 1, 2\}$ to find a lower bound of $\int_{-2}^{2} f(x)\, dx$.

We have

$$L(f, P) \leq \int_{-2}^{2} f(x)\, dx$$

We compute

$$L(f, P) = m_1(3 - 1) + m_2(4 - 3)$$

We have $m_1 = 0$ and $m_2 = 1$. And so

$$L(f, P) = 0(2) + 1(1) = 1.$$

Therefore

$$1 \leq \int_{-2}^{2} f(x)\, dx$$

Page 17 of 28 — add any extra pages after page 28 — Page 17 of 28

Question 13 (5 marks)

Let

$$f(x) = \begin{cases} 1 & x \in [0, 1/2) \\ 0 & x \in [1/2, 1]. \end{cases}$$

Prove $f(x)$ is integrable on $[0, 1]$.

Let $\epsilon > 0$. Let $\epsilon_1 = \epsilon_2 = \min\{\frac{1}{4}, \frac{\epsilon}{4}\}$. Consider the partition $P = \{0,\ 1/2 - \epsilon_1,\ 1 + \epsilon_2,\ 1\}$. Since $\epsilon_1 = \epsilon_2 < 1/2$, P is a partition. Notice

$$U(f, P) - L(f, P) = \epsilon_1 + \epsilon_2 \leq (\tfrac{\epsilon}{4} + \tfrac{\epsilon}{4}) = \tfrac{\epsilon}{2} < \epsilon.$$

Therefore for every $\epsilon > 0$ there exists a partition P such that $|U(f, P) - L(f, P)| < \epsilon$. Therefore by Theorem 6.14 from Theorem Sheets f is integrable.

Page 18 of 28 — add any extra pages after page 28 — Page 18 of 28

Question 14 (5 marks)

Find the interval of convergence of the following power series

$$\sum_{n=1}^{\infty} \frac{2^n}{n}(x-1)^n.$$

We apply the (limit) ratio test. Notice

$$\frac{\frac{2^{n+1}}{n+1}(x-1)^{n+1}}{\frac{2^n}{n}(x-1)^n} = \frac{n2(x-1)}{(n+1)}$$

For any $x \in \mathbb{R}$, the limit $\frac{n2(x-1)}{(n+1)} \to 2(x-1)$. If $|2(x-1)| < 1$, that is, $1/2 < x < 3/2$, the series converges.

If $|2(x-1)| > 1$, that is, $x < 1/2$ or $3/2 < x$, the series diverges.

If $x = 1/2$, the series becomes

$$\sum_{n=1}^{\infty} \frac{2^n}{n}(-\frac{1}{2})^n = \sum_{n=1}^{\infty} \frac{(-1)^n}{n}.$$

The series converges by alternating series test.

If $x = 3/2$, the series becomes

$$\sum_{n=1}^{\infty} \frac{2^n}{n}(\frac{1}{2})^n = \sum_{n=1}^{\infty} \frac{1}{n}.$$

This is the harmonic series and hence diverges.
The interval of convergence is $[1/2, 3/2)$.

Page 19 of 28 — add any extra pages after page 28 — Page 19 of 28

Question 15 (4 marks)

For each of the following functions, find the Taylor polynomial of order 2 around $x = 0$ if it exists, or explain why it does not exist as appropriate.

(a)

$$f(x) = \cos x$$

$T_2(x) = 1 - \frac{1}{2}x^2$

(b)

$$f(x) = \sqrt{x}\cos x$$

$f(x)$ is not differentiable at $x = 0$ so the 2nd order Taylor polynomial does not exist.

(c)

$$f(x) = \cos(x^2)$$

$T_2(x) = 1$

(d)

$$f(x) = \begin{cases} \cos x & x \leq 0 \\ 1 - x^2 & x > 0 \end{cases}$$

The second left derivatives at $x = 0$ is -1 while the second right derivate is -2, hence the second derivative does not exist at $x = 0$, neither does the 2nd order Taylor polynomial.

There is extra space for Question 15 on the subsequent page

Extra space for Question 15. Submit this page, even if left blank.

Page 20 of 28 — add any extra pages after page 28 — Page 20 of 28

Page 21 of 28 — add any extra pages after page 28 — Page 21 of 28

Question 16 (5 marks)

Let n be a non-negative integer. Let $C_n(\theta)$ denote the order n Taylor polynomial of $\cos(\theta)$ at 0. Prove

$$|C_n(\alpha) - \cos(\alpha)| < \frac{\alpha^{n+1}}{(n+1)!}$$

for all $\alpha \in \mathbb{R}$.

Let $\alpha \in (0, 2\pi)$. Let $n \in \mathbb{N}$. By Taylor's Theorem there exists $c \in (0, \alpha)$ where

$$\cos(\alpha) - C_n(\alpha) = \frac{f^{n+1}(c)}{(n+1)!}\alpha^{n+1}$$

We have

$$f^{n+1}(c) \in \{\sin(c), \cos(c), -\sin(c), -\cos(c)\}$$

Therefore

$$-1 \leq f^{n+1}(c) \leq 1$$

Therefore

$$-\frac{\alpha^{n+1}}{(n+1)!} \leq \cos(\alpha) - C_n(\alpha) \leq \frac{\alpha^{n+1}}{(n+1)!}$$

and so

$$|C_n(\alpha) - \cos(\alpha)| \leq \frac{\alpha^{n+1}}{(n+1)!}$$

Turn the page for appended material.

Do not submit appended material with exam submission.

End of Exam — Total Available Marks = 72